

Network Load Balancer (NLB)

Class No:22

Network Load Balancer (NLB)

The **AWS Network Load Balancer (NLB)** operates at the **transport layer (Layer 4)** of the OSI model and is designed to handle **millions of requests per second** with ultra-low latency. It is ideal for applications requiring high performance, scalability, and static IP addresses.

Key Features

- **Protocol Support:** NLB supports **TCP, UDP, TCP_UDP**, and **TLS** protocols, making it suitable for a wide range of use cases.
- **Static IP Addresses:** Each Availability Zone (AZ) can have a static IP or an Elastic IP, ensuring consistent endpoints for clients.
- **Cross-Zone Load Balancing:** Distributes traffic across all registered targets in enabled AZs, improving fault tolerance.
- **Health Checks:** Monitors the health of targets and routes traffic only to healthy ones.
- **Dynamic Scaling:** Automatically scales to handle traffic changes without manual intervention.
- **OSI Layer:** Operates at **Layer 4 (Transport Layer)**, focusing on TCP/UDP traffic rather than HTTP headers.
- **Static IP Support:** Provides a **single static IP per Availability Zone**, making it suitable for applications that require fixed IPs.
- **High Performance:** Designed to handle **millions of requests per second** with ultra-low latency.
- **Port-Based Routing:** Routes traffic based on **port numbers**, enabling flexible backend service mapping.

Components

- **Load Balancer:** Acts as the single point of contact for clients and distributes traffic to targets.
- **Listeners:** Configured with protocols and ports to check for client connection requests.
- **Target Groups:** Define the targets (e.g., EC2 instances, IP addresses) and health check settings
- **Key Differentiator vs. ALB:**
 - ALB works at Layer 7 (application-level, HTTP/HTTPS).
 - NLB works at Layer 4, making it better for raw transport traffic and non-web workloads.

Quick Comparison Table

Feature	Network Load Balancer (NLB)	Application Load Balancer (ALB)
OSI Layer	Layer 4 (Transport)	Layer 7 (Application)
Protocols Supported	TCP, UDP, TLS	HTTP, HTTPS, gRPC
Static IP	✓ Yes	✗ No
Performance	Millions of requests/sec	High, but optimized for HTTP
Routing Type	Port-based	Path-based, host-based
Best Use Case	Gaming, IoT, backend apps	Web apps, microservices

Hands on: Creating a Network load balancer with two EC2

Step no:1 Create two EC2 instances,

EC2 no:1

Name: **My-target-instance-1**

Select existing group → default security group → default VPC → launch instance

EC2 no:2

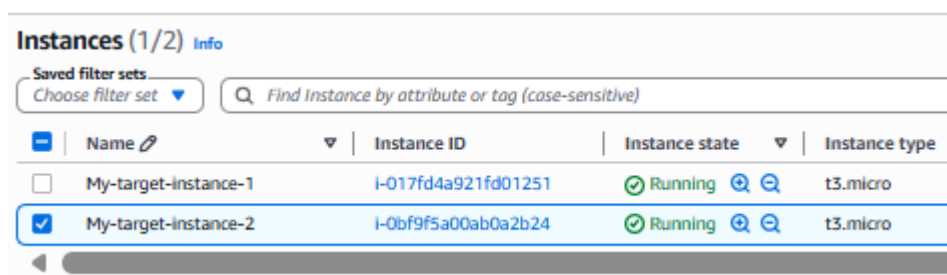
Name: **My-target-instance-2**

Select existing group → default security group → default VPC → launch instance

In security group,

Ensure **All TCP** is enabled.

Use the same key pair for both.



The screenshot shows the AWS Management Console 'Instances' page. At the top, it says 'Instances (1/2)' with an 'Info' link. Below this is a 'Saved filter sets' section with a dropdown menu and a search bar labeled 'Find Instance by attribute or tag (case-sensitive)'. The main table lists two instances:

	Name	Instance ID	Instance state	Instance type
☐	My-target-instance-1	i-017fd4a921fd01251	Running	t3.micro
☒	My-target-instance-2	i-0bf9f5a00ab0a2b24	Running	t3.micro

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

My-target-instance-1

[Add additional tags](#)

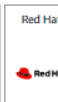
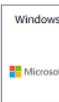
▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

[Quick Start](#)



[Browse more AMIs](#)

Including AMIs from AWS, Marketplace, and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-048f445314bcaa09 (64-bit (x86), uefi-preferred) / ami-07efda20bfc979493 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260406.2 x86_64 HVM kernel-6.1

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-048f445314bcaa09

Publish Date

2026-04-06

Username

ec2-user



Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t5.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0112 USD per Hour
On-Demand SUSE base pricing: 0.0112 USD per Hour On-Demand Windows base pricing: 0.0204 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour

Free tier eligible

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

mumkp

[Create new key pair](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

mumkp

[Create new key pair](#)

▼ Network settings [Info](#)

[Edit](#)

[Network](#) | [Info](#)

vpc-0538c231a226fe3c9

[Subnet](#) | [Info](#)

No preference (Default subnet in any availability zone)

[Auto-assign public IP](#) | [Info](#)

☐ Enable

[Firewall \(security groups\)](#) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

[Common security groups](#) | [Info](#)

Select security groups

default sg-06d80e515c32b9515
VPC: vpc-0538c231a226fe3c9

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Step no:2 Launch the My-target-instance-1

[illegible]

i-017fd4a921fd01251 (My-target-instance-1)

PublicIPs: 3.110.173.99 PrivateIPs: 172.31.42.76

Enter the commands:

Sudo su → switch user

yum install httpd → install the httpd server

```

Last login: Thu Apr  9 13:21:57 2026 from 13.233.177.5
[ec2-user@ip-172-31-42-76 ~]$ sudo su
[root@ip-172-31-42-76 ec2-user]# yum install httpd
Last metadata expiration check: 0:01:05 ago on Thu Apr  9 13:23:04 2026.
Dependencies resolved.
=====
Package                                     Architecture
=====
Installing:
httpd                                       x86_64
Installing dependencies:
apr                                         x86_64
apr-util                                   x86_64
apr-util-ldap                              x86_64
generic-logos-httpd                       noarch
httpd-core                                 x86_64
httpd-filesystem                          noarch
httpd-tools                               x86_64
libbrotli                                 x86_64

```

i-017fd4a921fd01251 (My-target-instance-1)

PublicIPs: 3.110.173.99 PrivateIPs: 172.31.42.76

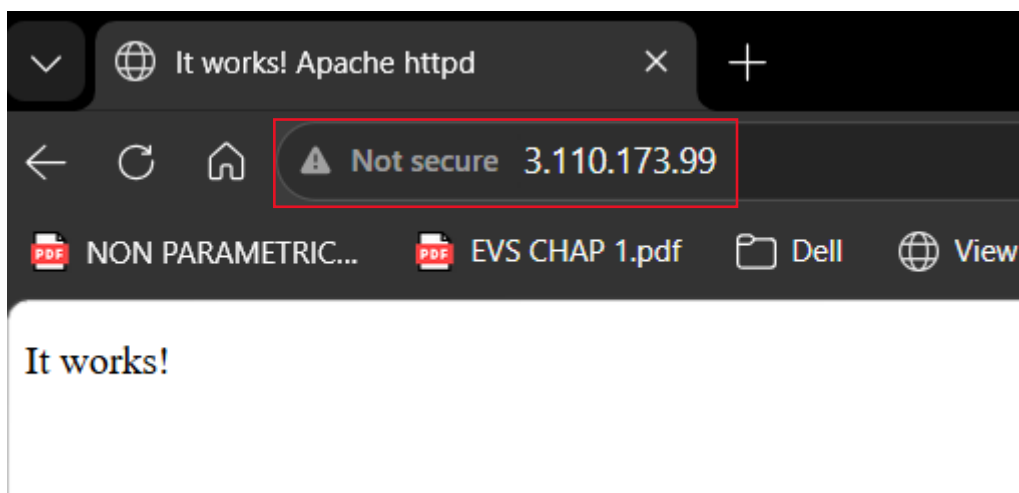
Systemctl start httpd → start the httpd server

```
[root@ip-172-31-42-76 ec2-user]# systemctl start httpd
```

i-017fd4a921fd01251 (My-target-instance-1)

PublicIPs: 3.110.173.99 PrivateIPs: 172.31.42.76

To check it's working → copy the **public ip address** → paste it in a new private browser → we can see the output



Step no:3 Launch the My-target-instance-2

```

#
~\  ### Amazon Linux 2023
~~\  #####
~~\  #####|
~~\  \#/ https://aws.amazon.com/linux/ama
~~  V~' '->
~~~~
~~~~
~~~.~.
~~~/_/
[ec2-user@ip-172-31-38-0 ~]$

```

i-0bf9f5a00ab0a2b24 (My-target-instance-2)

PublicIPs: 65.1.86.173 PrivateIPs: 172.31.38.0

Enter the commands:

Sudo su → switch user

yum install nginx → install the nginx server

```

_/_/m/'
[ec2-user@ip-172-31-38-0 ~]$ sudo su
[root@ip-172-31-38-0 ec2-user]# yum install nginx
Amazon Linux 2023 Kernel Livepatch repository
Amazon Linux 2023 Kernel Livepatch repository
00:00
Dependencies resolved.

=====
Package                               Architect
Size
=====
Installing:
  nginx                               x86_64
    33 k
Installing dependencies:

```

i-0bf9f5a00ab0a2b24 (My-target-instance-2)

PublicIPs: 65.1.86.173 PrivateIPs: 172.31.38.0

Systemctl start nginx → start the nginx server

```

[root@ip-172-31-38-0 ec2-user]# systemctl start nginx
[root@ip-172-31-38-0 ec2-user]# systemctl status nginx
• nginx.service - The nginx HTTP and reverse proxy server
  Loaded: loaded (/usr/lib/systemd/system/nginx.service; vendor preset: enabled)
  Active: active (running) since Thu 2026-04-09 13:32:24 UTC; 1min 1s ago
  Process: 25759 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
  Process: 25776 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
  Process: 25817 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
  Main PID: 25847 (nginx)
  Tasks: 3 (limit: 1014)
  Memory: 3.4M
  CPU: 71ms
  CGroup: /system.slice/nginx.service
          └─25847 "nginx: master process /usr/sbin/nginx"
             └─25848 "nginx: worker process"
                └─25849 "nginx: worker process"

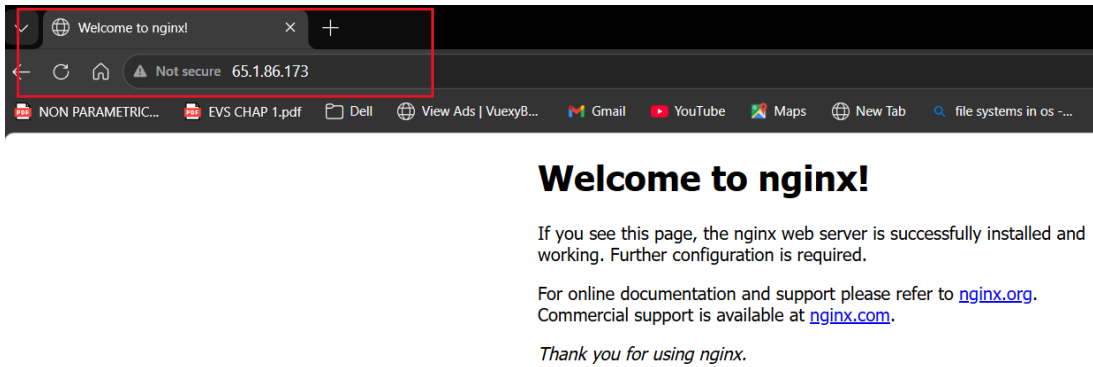
Apr 09 13:32:24 ip-172-31-38-0.ap-south-1.compute.internal:~$
Apr 09 13:32:24 ip-172-31-38-0.ap-south-1.compute.internal:~$
Apr 09 13:32:24 ip-172-31-38-0.ap-south-1.compute.internal:~$

```

i-0bf9f5a00ab0a2b24 (My-target-instance-2)

PublicIPs: 65.1.86.173 PrivateIPs: 172.31.38.0

To check it's working → copy the **public ip address** → paste it in a new private browser → we can see the output



Step no:4 By default both the httpd and nginx server will listen to default port 80.If we didn't change it and used them in load balancer it will show as error.

We have to change any one of their or both their port number.

For EC2 instance -1:

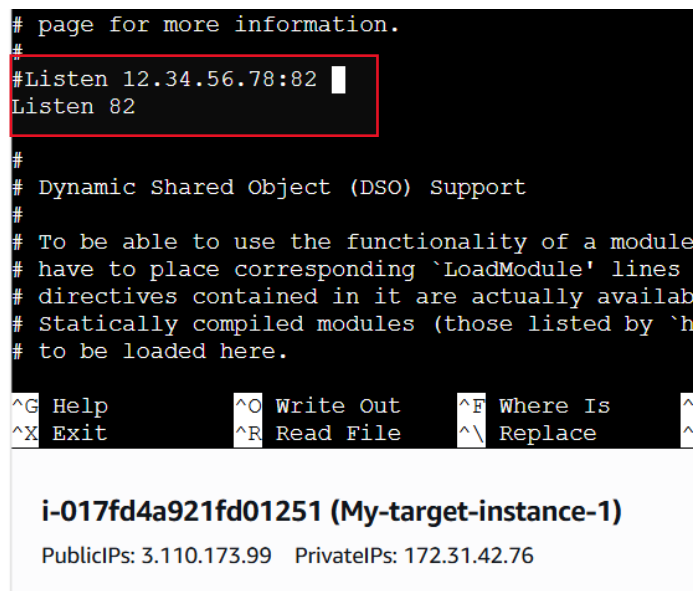
We are going to change the port number as 82.

Launch the instance → connect → Enter the command: **nano /etc/httpd/conf/httpd.conf**

This is where the server configuration is located.

```
root@ip-172-31-42-76 ec2-user]# nano /etc/httpd/conf/httpd.conf
root@ip-172-31-42-76 ec2-user]#
```

The file will be opened → Scroll down → On listen → change port number as 82 → cntrl+x and cntl+y → enter



Restart the httpd server: **systemctl restart httpd**

```
complete!  
[root@ip-172-31-42-76 ec2-user]# systemctl restart httpd  
  
[root@ip-172-31-42-76 ec2-user]#  
[root@ip-172-31-42-76 ec2-user]#
```

i-017fd4a921fd01251 (My-target-instance-1)

PublicIPs: 3.110.173.99 PrivateIPs: 172.31.42.76

For EC2 instance -2:

We are going to change the port number as 84.

Launch the instance → connect → Enter the command: **nano /etc/nginx/nginx.conf**

This is where the server configuration is located.

```
[root@ip-172-31-38-0 ec2-user]# nano /etc/nginx/nginx.conf  
[root@ip-172-31-38-0 ec2-user]#
```

The file will be opened → Scroll down → On server → change port number as 84 → cntrl+x and cntl+y → enter

```
# Load modular configuration files from the /etc/nginx/conf.d directory  
# See http://nginx.org/en/docs/nginx_core_module.html#include  
# for more information.  
include /etc/nginx/conf.d/*.conf;  
  
server {  
    listen      84;  
    listen      [::]:84;  
    server_name _;  
    root        /usr/share/nginx/html;  
  
    # Load configuration files for the default server block.  
    include /etc/nginx/default.d/*.conf;  
  
    error_page 404 /404.html;  
    location = /404.html {  
        # ...  
    }  
}
```

G Help ^O Write Out ^F Where Is ^R Cut ^T Exe
X Exit ^R Read File ^\ Replace ^U Paste ^J Jus

i-0bf9f5a00ab0a2b24 (My-target-instance-2)

PublicIPs: 65.1.86.173 PrivateIPs: 172.31.38.0

Restart the nginx server: **systemctl restart nginx**

```
[root@ip-172-31-38-0 ec2-user]# systemctl restart nginx
[root@ip-172-31-38-0 ec2-user]#
```

i-0bf9f5a00ab0a2b24 (My-target-instance-2)

PublicIPs: 65.1.86.173 PrivateIPs: 172.31.38.0

Step no:5 Create two security groups for the two respective instances.

For EC2-1,

Target type: **Instances**

Name: **My-Tg-1**

Protocol: **TCP**

Port: **82**

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation."

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: **ALB** **NLB** **GWLB**

☐ IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: **ALB** **NLB** **GWLB**

☐ Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: **ALB**

☐ Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: **NLB**

Target group name

Name must be unique per Region per AWS account.

My-Tg-1

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 7/32

Protocol

Protocol for communication between the load balancer and targets.

TCP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

82

1-65535

VPC: default VPC

Health checks: **TCP** (if we didn't choose this, the NLB will not work)

Click: **Next**

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0538c231a226fe3c9 (default)  [Create VPC](#)

Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol
TCP

► **Advanced health check settings**

Attributes

① Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

► **Tags - optional**
Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Cancel](#) [Next](#)

Choose the instance 1 → include as pending below → next

Available instances (1/2)

☐	Instance ID	Name	State	Security groups	Zone
☒	i-017fd4a921fd01251	My-target-instance-1	Running	default	ap-south-1a
☐	i-0bf9f5a00ab0a2b24	My-target-instance-2	Running	default	ap-south-1a

1 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

82

1-65535 (separate multiple ports with commas)

[Include as pending below](#)

Click **Create target group**.

Review and create
Review your target group configuration before creating

Step 1: Target group details [Edit](#)

Target group details

Name My-Tg-1	Target type Instance	Protocol : Port TCP: 82	Protocol version HTTP1
VPC vpc-0538c231a226fe3c9	IP address type IPv4		

Health check details

Health check protocol TCP	Health check port traffic-port	Interval 30 seconds	Timeout 10 seconds
Healthy threshold 5	Unhealthy threshold 2		

Step 2: Register targets [Edit](#)

Targets (1)

Instance ID	Name	Port	Zone
i-017fd4a921fd01251	My-target-instance-1	82	ap-south-1a

[Cancel](#) [Previous](#) [Create target group](#)

For EC2-2,

Target type: **Instances**

Name: **My-Tg-2**

Protocol: **TCP**

Port: **84**

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health

Settings - *immutable*

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: **ALB** **NLB** **GWLB**

☐ IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: **ALB** **NLB** **GWLB**

☐ Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: **ALB**

☐ Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: **NLB**

Target group name

Name must be unique per Region per AWS account.

My-Tg-2

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 7/32

Protocol

Protocol for communication between the load balancer and targets.

TCP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

84

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

VPC: **default VPC**

Health checks: **TCP** (if we didn't choose this, the NLB will not work)

Click: **Next**

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0538c231a226fe3c9
172.31.0.0/16

(default)



Create VPC

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

TCP

Advanced health check settings

Attributes

Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.

Tags - optional

Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.

Cancel

Next

Choose the instance 2 → include as pending below → next

Instance ID

Name

State

Security groups

Zone

Private IPv4 address

Subnet ID

☐

i-017fd4a921fd01251

My-target-instance-1

Running

default

ap-south-1a

172.31.42.76

subnet-01bf9f5a00ab0a2b24

☒

i-0bf9f5a00ab0a2b24

My-target-instance-2

Running

default

ap-south-1a

172.31.38.0

subnet-01bf9f5a00ab0a2b24

1 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

84

1-65535 (separate multiple ports with commas)

Include as pending below

1 selection is now pending below. Include more or register targets when ready.

Review targets

Targets (1)

Filter targets

Show only pending

Remove all pending

Instance ID

Name

Port

State

Security groups

Zone

Private IPv4 address

Subnet ID

Launch time

[i-0bf9f5a00ab0a2b24](#)

My-target-instance-2

84

Running

default

ap-south-1a

172.31.38.0

subnet-01bf9f5a00ab0a2b24

April 9, 2026, 18:45 (UTC+05:30)

1 pending

Cancel

Previous

Next

Click Create target group.

Review and create

Review your target group configuration before creating

Step 1: Target group details

Edit

Target group details

Name

My-Tg-2

Target type

Instance

Protocol : Port

TCP: 84

Protocol version

HTTP1

VPC

[vpc-0538c231a226fe3c9](#)

IP address type

IPv4

Health check details

Health check protocol

TCP

Health check port

traffic-port

Interval

30 seconds

Timeout

10 seconds

Healthy threshold

5

Unhealthy threshold

2

Step 2: Register targets

Edit

Targets (1)

Instance ID

Name

Port

Zone

[i-0bf9f5a00ab0a2b24](#)

My-target-instance-2

84

ap-south-1a

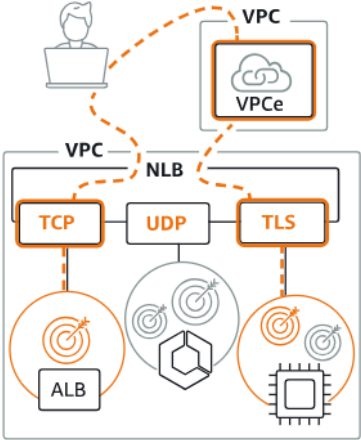
Cancel

Previous

Create target group

Step no:6 Create Load Balancer → choose NLB.

Network Load Balancer [Info](#)



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

[Create](#)

ion

Name: **My-Nlb** → internet facing → default VPC

[Load balancers](#) > Create Network Load Balancer

► How Network Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

My-Nlb

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves Internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.

Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to on-premises targets or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0538c231a226fe3c9

172.31.0.0/16

(default) ▼

Network mapping,

Choose **all AZs** to make it availability high.

Network mapping [info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing is configured. To confirm the VPC for your targets, view [target groups](#).

vpc-0538c231a226fe3c9
172.31.0.0/16

Availability Zones and subnets

Select one or more Availability Zones and corresponding subnets. Enabling multiple Availability Zones increases the fault tolerance of your applications. If you select multiple Availability Zones, the load balancer and the VPC are not available for selection.

☒ **ap-south-1a (aps1-az1)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to be configured to allow traffic.

subnet-01bf4d440c0912db3
172.31.32.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

IPv4 address

The front-end IPv4 address of the load balancer in the selected Availability Zone.

☐ Assigned by AWS

☒ **ap-south-1b (aps1-az3)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to be configured to allow traffic.

subnet-0b3c21b6de4314d95
172.31.0.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

IPv4 address

The front-end IPv4 address of the load balancer in the selected Availability Zone.

☐ Assigned by AWS

☒ **ap-south-1c (aps1-az2)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to be configured to allow traffic.

Security group,

Choose **default security group**

Listeners and routing,

- Choose: **TCP** → Port: **82**
- Select the target group: **My-tg-1** (belongs to EC2-1)

Security groups [info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected for your load balancer.

Security groups - recommended

Security groups support on Network Load Balancers can only be enabled at creation by including at least one security group. You can change security groups after creation for PrivateLink Network Load Balancers, security group rules are enforced on PrivateLink traffic; however, you can turn off inbound rules for PrivateLink Network Load Balancers.

Select up to 5 security groups

default (sg-06d80e515c32b9515) X
Inbound rule (0) Outbound rule (1)

Listeners and routing [info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how traffic is routed to your targets.

▼ Listener TCP:82

Protocol TCP Port 82
1-65535

Forward to target group [info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group My-Tg-1 TCP Weight 1
Target type: Instance, IPv4 | Target stickiness: Off 0-999

[Add target group](#)

You can add up to 4 more target groups.

Target group stickiness [info](#)

Enables the load balancer to bind a user's session to a specific target group. If you want to bind a user's session to a specific target, turn on the Target Group attribute.

☐ Turn on target group stickiness

Weighted routing evaluation - recommended

Your Network Load Balancer's ability to distribute requests according to assigned target group weights can be hindered if your target groups don't follow best practices.

[Evaluate](#)

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add listener tag](#)

You can add up to 50 more tags.

Add listener

- Choose: **TCP** → Port: **84**
- Select the target group: **My-tg-2** (belongs to EC2-2)

▼ Listener TCP:84

Protocol: TCP Port: 84 (1-65535)

Forward to target group | Info
Choose a target group and specify routing weight or [create target group](#).

Target group: My-Tg-2 (Target type: Instance, IPv4 | Target stickiness: Off) TCP Weight: 1 (0-999) Percent: 100%

[Add target group](#)
You can add up to 4 more target groups.

Target group stickiness | Info
Enables the load balancer to bind a user's session to a specific target group. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.
☐ Turn on target group stickiness

Weighted routing evaluation - recommended
Your Network Load Balancer's ability to distribute requests according to assigned target group weights can be hindered if your target groups don't follow best practices.
[Evaluate](#)

Listener tags - optional
Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.
[Add listener tag](#)
You can add up to 50 more tags.

[Add listener](#)
You can add up to 48 more listeners.

Review and click → Create Load balancer.

Review

Review the load balancer configurations and make changes if needed. After you finish reviewing the configurations, choose Create load balancer.

Summary

Review and confirm your configurations. [Estimate cost](#)

Basic configuration [Edit](#)

Name: My-Nlb
Scheme: Internet-facing
IP address type: IPv4

Network mapping [Edit](#)

VPC: [vpc-0538c231a226fe3c9](#)
Availability Zones and subnets:

- ap-south-1a
[subnet-01bf4d440c0912db3](#)
- ap-south-1b
[subnet-0b3c21b6de4314d95](#)
- ap-south-1c
[subnet-0c305b80f72541592](#)

Security groups [Edit](#)

default
[sg-06d80e515c32b9515](#)

Listeners and routing [Edit](#)

TCP:82 | Forward to 1 target group
TCP:84 | Forward to 1 target group

Service integrations [Edit](#)

AWS Global Accelerator: -

Tags [Edit](#)

-

Attributes

[i](#) Certain default attributes will be applied to your load balancer. You can view and edit them after creating the load balancer.

Creation workflow and status

► Server-side tasks and status

After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

[Cancel](#)

[Create load balancer](#)

Once created wait for the **status** to become **active**.

Copy the **DNS** of the NLB.

My-NLB

Actions

Details

Load balancer type

Network

Scheme

Internet-facing

Status

Active

Hosted zone

ZVDDRBQ08TROA

VPC

vpc-0538c231a226fe3c9

Availability Zones

subnet-01bf4d440c0912db3 ap-south-1a (aps1-az1)
subnet-0c305b80f72541592 ap-south-1c (aps1-az2)
subnet-0b3c21b6de4314d95 ap-south-1b (aps1-az3)

Load balancer IP address type

IPv4

Date created

April 9, 2026, 19:39 (UTC+05:30)

Load balancer ARN

arn:aws:elasticloadbalancing:ap-south-1:314383175192:loadbalancer/net/My-NLB/26e5e453f6aa3c17

DNS name

My-NLB-26e5e453f6aa3c17.elb.ap-south-1.amazonaws.com (A Record)

Listeners

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Tags

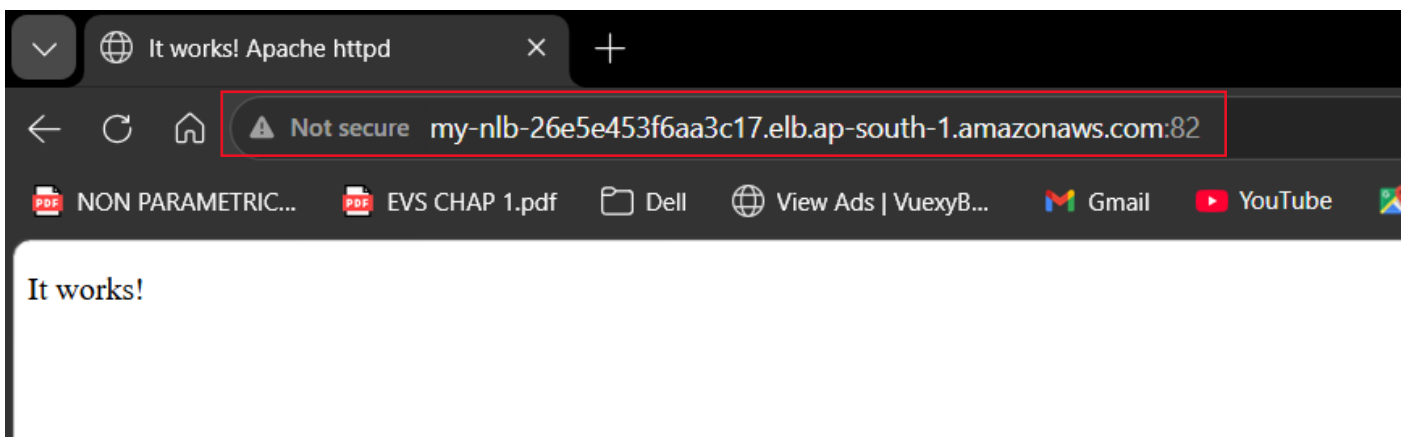
Listeners (2)

A listener checks for connection requests using the protocol and port that you configure. Traffic received by a Network Load Balancer listener is forwarded to the selected target group.

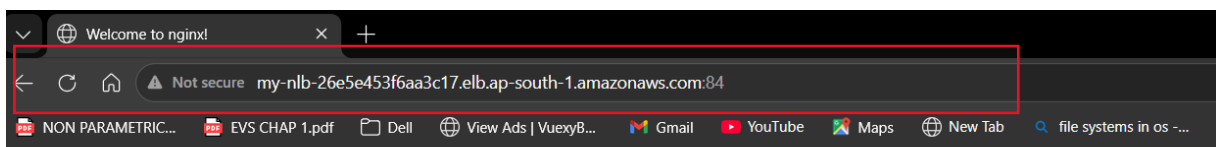
Filter listeners

☐	Protocol:Port	Default action	ARN	Security policy	Default SSL/TLS certificate	ALPN policy	Tags
☐	TCP:84	Forward to target group My-Tg-2 (100%) Target group stickiness: Off	ARN	Not applicable	Not applicable	Not applicable	0 tags
☐	TCP:82	Forward to target group My-Tg-1 (100%) Target group stickiness: Off	ARN	Not applicable	Not applicable	Not applicable	0 tags

Paste the **DNS:82** in a private browser to check the **httpd** is working.



Paste the **DNS:84** in a private browser to check the **nginx** is working.



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

If it shows the respective output, it means the NLB is working fine.